MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/206-00018 and M17/206-00030

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-17.

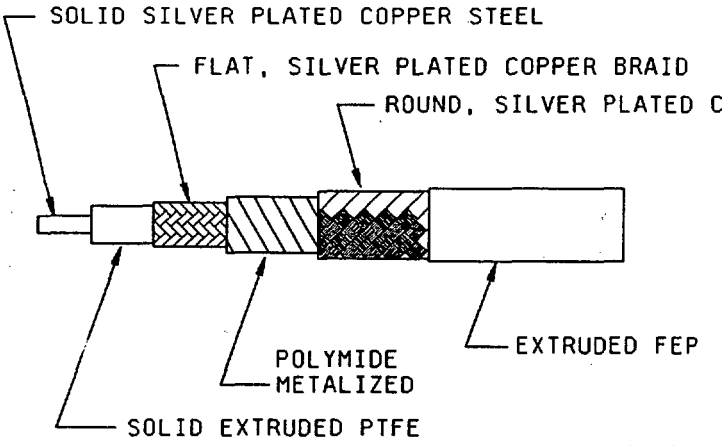


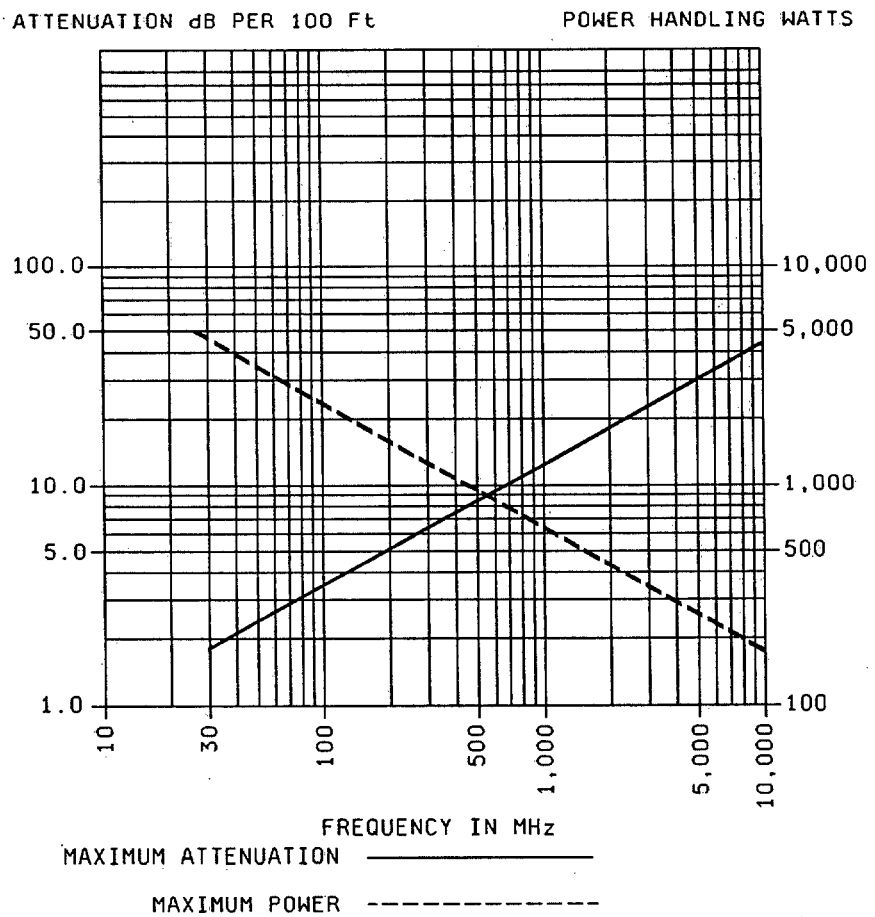
FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid silver coated copper wire, Overall diameter: .00365 inch ±.0002.
Dielectric core	Type F-1: Solid PTFE. Diameter: .117 inch ±.003.
Outer conductor Inner shield helical wrap	Double braid with meadalized polyimide interlayer. Diameter: .153 inch ±.005. Carriers: 16 Ends: N/A Picks/inch: 18-19 Wire type: Flat, silver plated copper wire. .030 inch ±.003 x .0020 inch ±.0005.
Outer braid	Carriers: 24 Ends: 4 Picks/inch: 23-25 Wire type: AWG 36, silver plated copper wire. Diameter: .153 inch ±.005.
Jacket	Type IX extruded FEP Diameter: .169 inch ±.005.

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ATTENUATION	
MHz	dB
50	2.5
400	7.4
3,000	22.2
11,000	47.8

FIGURE 2. Power rating and attenuation at 25°C sea level.

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ENGINEERING INFORMATION:

Continuous working voltage: 1,400 V rms, maximum.

Operating frequency: 30 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 19.5 ohms per 1000 feet.

Elongation: 1 percent, minimum.

Engineering notes: This cable is useful in general purpose, high temperature applications (connector series 2.4 mm 3.5 mm and SMK).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Out-of-roundness: 10%, maximum.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 4 lbs. minimum, 30 lbs, maximum.

Aging stability: Not applicable.

Cold bend: -55°C ±2°C.

Stress crack resistance: ±230°

Dimensional stability: +200°C ±5°C.

Inner conductor from core: .187 inch, maximum.

Inner conductor from jacket: .312 inch, maximum.

Contamination: Not applicable.

Flame propagation: Applicable.

Acid gas generation: Not applicable.

Halogen content: Not applicable.

Immersion test: Not applicable.

Smoke index: Not applicable.

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Toxicity index: Not applicable.

Durometer hardness: Not applicable.

Weathering: Not applicable.

Abrasion resistance: Not applicable.

Tear strength: Not applicable.

Heat distortion: Not applicable.

Physical tests on unaged jacket: Not applicable.

Physical tests on aged jacket: Not applicable.

Hot oil immersion: Not applicable.

Weight: 4 pounds per 100 feet maximum.

Electrical:

Test frequency: 50 MHz to 18 GHz, M17/206-00018
50 MHz to 30 GHz, M17/206-00030

Spark test: 2,000 V rms, + 25%, -0%.

Voltage withstanding: 3,000 V rms, minimum.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,900 V rms, minimum.

Characteristic impedance: 50 ±2 ohms.

Attenuation: See figure 2.

Structural return loss: 1.15:1 at 12 GHz, 1.25:1 at 18 GHz, 1.30:1 at 30 GHz, maximum.

Capacitance: 32 pF per foot.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise: Not applicable.

Time delay: Not applicable.

Part or Identifying Number (PIN): M17/206-00018 and M17/206-00030.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, ME, MI
Navy - AS, MC, OS, SH
Air Force - 19, 80, 99
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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